

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Open Data Analysis Project

COURSE NAME: Query Processing for Data Science

COURSE CODE: DSA0503

COURSE OUTCOME COVERED

CO4: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.

Assessment Tool Weightage: 40%

Total Marks: 20

Code of Conduct

I V.Tejaswini(192424360) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Open Data Analysis Project Rubric

Data Collection & Understanding (10 Marks)	Data Cleaning & Preparation (10 Marks)	Exploratory Data Analysis (10 Marks)	Visualization & Dashboard Design (10 Marks)	Demonstration (10 Marks)	Total (50 Marks)

Total Marks Awarded:

Signature of the Course Faculty

Questions:

1. Road Accident Analysis and Traffic Safety Evaluation

Using open road accident datasets from Kaggle Datasets or government transportation portals, investigate factors contributing to traffic accidents. Analyze accident frequency with respect to time, weather conditions, road types, and geographic locations. Apply data cleaning techniques, identify high-risk zones, create visual dashboards, and provide recommendations for improving road safety and reducing accident rates based on the insights obtained.

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

# Load dataset

try:

    df = pd.read_csv("road_accidents.csv")

except FileNotFoundError:

    print("road_accidents.csv not found. Creating a dummy DataFrame for demonstration.")

    data = {

        'Weather': ['Clear', 'Rain', 'Clear', 'Fog', 'Clear', 'Rain'],

        'Road_Type': ['Motorway', 'Urban', 'Rural', 'Urban', 'Motorway', 'Rural'],

        'Location': ['Location A', 'Location B', 'Location A', 'Location C', 'Location B', 'Location A'],

        'Time': ['10:00:00', '15:30:00', '08:00:00', '22:00:00', '14:00:00', '07:00:00'],

        'Severity': [1, 2, 1, 3, 2, 1],

        'Number_of_Vehicles': [2, 3, 1, 2, 4, 1]

    }

    df = pd.DataFrame(data)

# Display dataset

print(df.head())

# Data cleaning
```

```
print("Missing values:")
print(df.isnull().sum())
df = df.drop_duplicates()
df = df.ffill() # Updated to use the recommended ffill() method
# Accident frequency by weather
weather = df["Weather"].value_counts()
plt.figure(figsize=(8,5))
weather.plot(kind="bar")
plt.title("Accidents by Weather Condition")
plt.xlabel("Weather Condition")
plt.ylabel("Number of Accidents")
plt.show()
# Accident frequency by road type
road = df["Road_Type"].value_counts()
plt.figure(figsize=(8,5))
road.plot(kind="bar")
plt.title("Accidents by Road Type")
plt.xlabel("Road Type")
plt.ylabel("Number of Accidents")
plt.show()
# Accident frequency by location
location = df["Location"].value_counts().head(10)
plt.figure(figsize=(10,5))
location.plot(kind="bar")
plt.title("Top 10 Accident-Prone Locations")
plt.xlabel("Location")
plt.ylabel("Number of Accidents")
plt.xticks(rotation=45)
plt.show()
```

```

# Accident frequency by time
df["Time"] = pd.to_datetime(df["Time"], errors="coerce")
df["Hour"] = df["Time"].dt.hour
hour = df["Hour"].value_counts().sort_index()
plt.figure(figsize=(10,5))
plt.plot(hour.index, hour.values, marker="o")
plt.title("Accidents by Time of Day")
plt.xlabel("Hour")
plt.title("Accidents by Time of Day")
plt.xlabel("Hour")
plt.ylabel("Number of Accidents")
plt.grid()
plt.show()

# Correlation analysis
numeric_data = df.select_dtypes(include="number")
plt.figure(figsize=(10,6))
sns.heatmap(numeric_data.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Between Accident Factors")
plt.show()

# High-risk locations
print("\nHigh-Risk Accident Zones:")
print(location)

# Recommendations
print("\nRecommendations:")
print("1. Improve road infrastructure in high-risk zones.")
print("2. Increase traffic monitoring during peak accident hours.")
print("3. Install proper road signs and street lighting.")

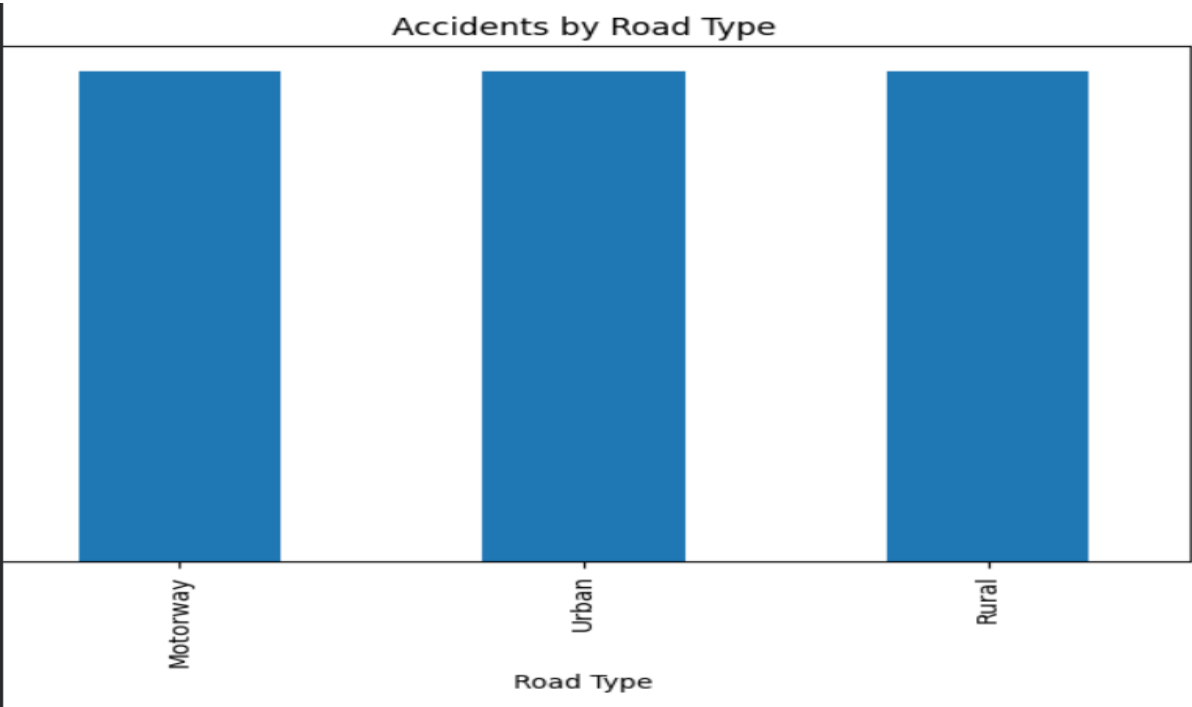
```

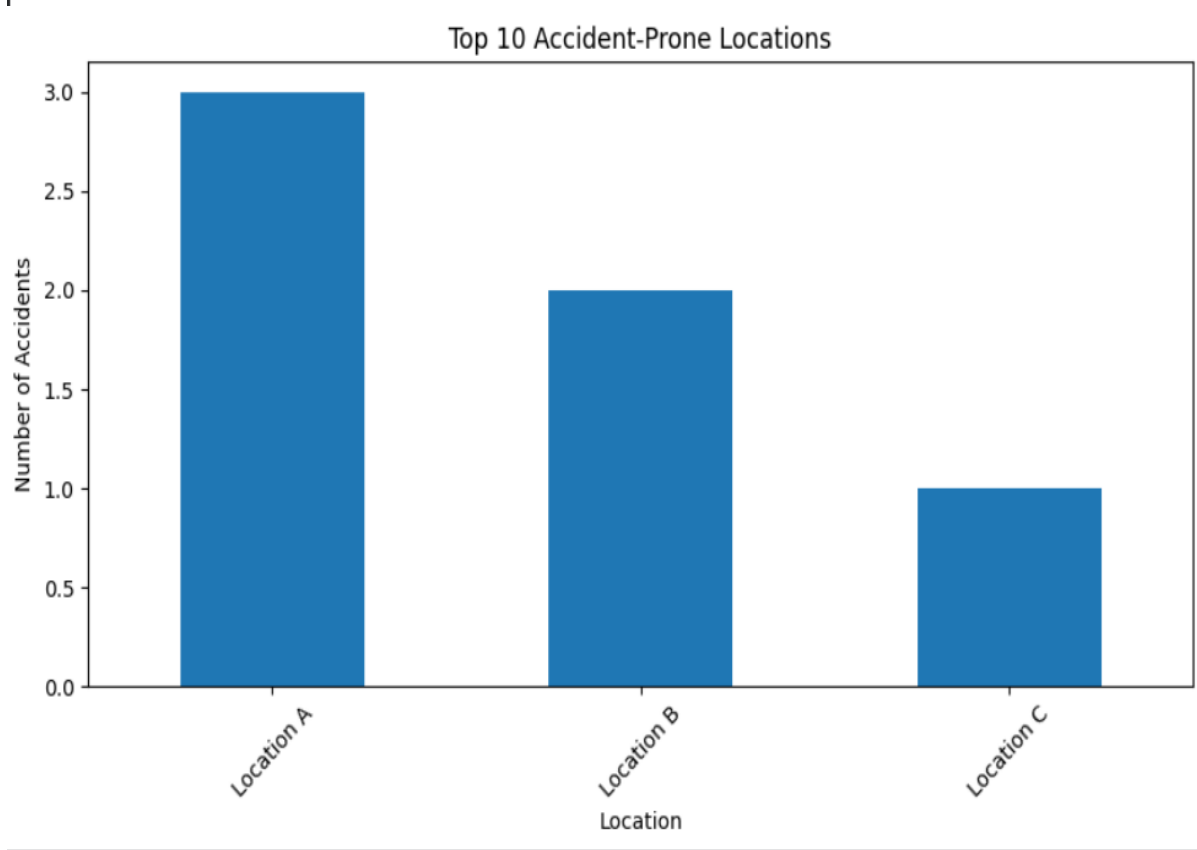
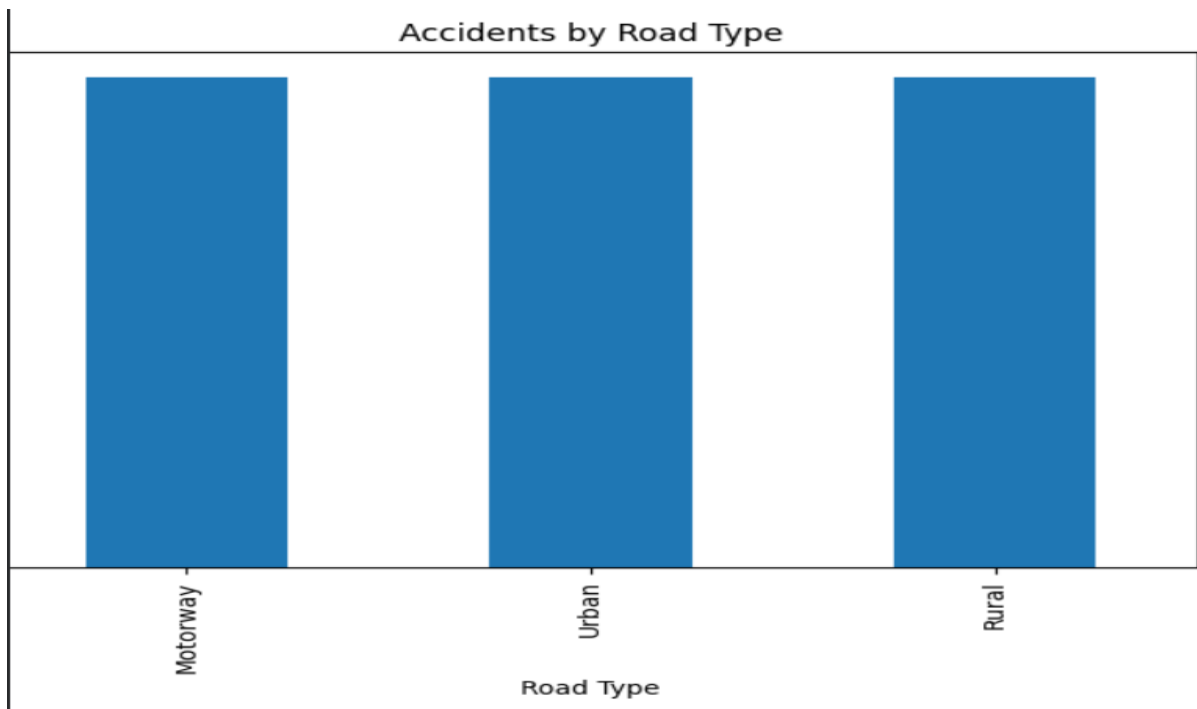
print("4. Take additional safety measures during bad weather.")

print("5. Enforce speed limits in accident-prone areas.")

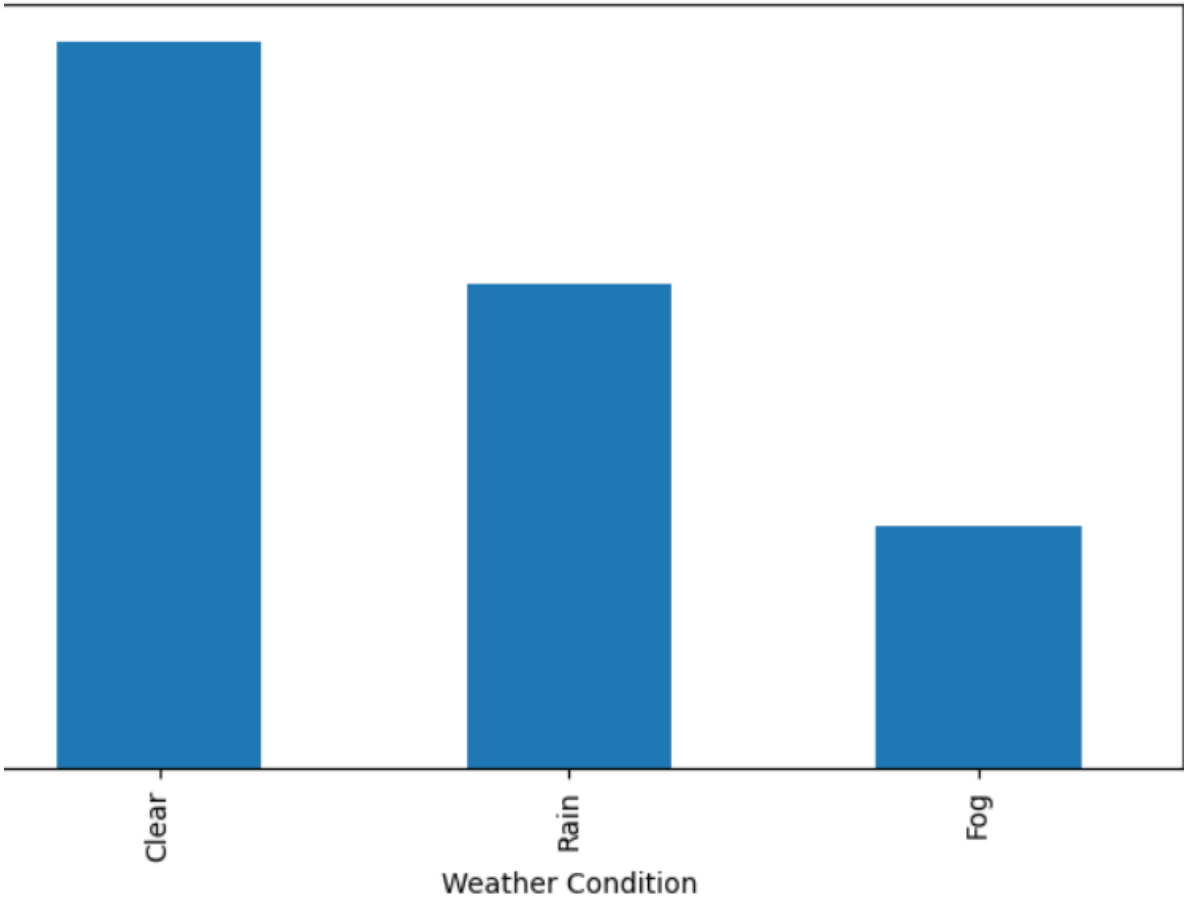
output:

```
nts.csv not found. Creating a dummy DataFrame for demonstration.
Road_Type  Location    Time  Severity  Number_of_Vehicles
Motorway    Location A  10:00:00      1           2
Urban       Location B  15:30:00      2           3
Rural       Location A   8:00:00      1           1
Urban       Location C  22:00:00      3           2
Motorway    Location B  14:00:00      2           4
ues:
      0
      0
      0
      0
      0
      0
ehicles  0
4
In file _5466/4227597926.py:28: FutureWarning: DataFrame.fillna with 'method' is deprecated and will raise
fillna(method="ffill")
```





Accidents by Weather Condition



Accidents by Road Type

